

Defining the process and cpu scheduler classes

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In [19]: class Process:
def __init__(self, pid, burst_time, arrival_time):
    self.pid = pid
    self.burst_time = burst_time
    self.remaining_time = burst_time
    self.arrival_time = arrival_time
    self.completion_time = None
    self.waiting_time = None

def execute(self, quantum):
    if self.remaining_time > quantum:
        self.remaining_time -= quantum
        return quantum
    else:
        execution_time = self.remaining_time
        self.remaining_time = 0
        return execution_time

def complete(self, completion_time):
    self.completion_time = completion_time
    self.waiting_time = self.completion_time - self.burst_time - self.arrival_time

class RoundRobinScheduler:
def __init__(self, processes, quantum):
    self.processes = processes
    self.quantum = quantum
    self.current_time = 0

def schedule(self):
    ready_queue = self.processes.copy()
    while ready_queue:
        current_process = ready_queue.pop(0)
        execution_time = current_process.execute(self.quantum)
        self.current_time += execution_time
        if current_process.remaining_time > 0:
            ready_queue.append(current_process)
        else:
            current_process.complete(self.current_time)
            print(f"Process {current_process.pid} completed at time {self.current_time}")

    total_waiting_time = sum(process.waiting_time for process in self.processes)
    average_waiting_time = total_waiting_time / len(self.processes)
    print(f"Average waiting time: {average_waiting_time}")
```

Applying the algorithm

Process ID	Arrival Time	Burst Time
1	0	5
2	1	6
3	2	3
4	3	1
5	4	5
6	6	4

```
In [20]: # Create a list of processes with arrival times
processes = [
    Process(1, 5, 0),
    Process(2, 6, 1),
    Process(3, 3, 2),
    Process(4, 1, 3),
    Process(5, 5, 4),
    Process(6, 4, 6),
]

# Create a Round Robin scheduler with a quantum of 4
scheduler = RoundRobinScheduler(processes, 4)

# Schedule the processes
scheduler.schedule()

Process 3 completed at time 11
Process 4 completed at time 12
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Process 6 completed at time 20
Process 1 completed at time 21
Process 2 completed at time 23
Process 5 completed at time 24
Average waiting time: 11.833333333333334